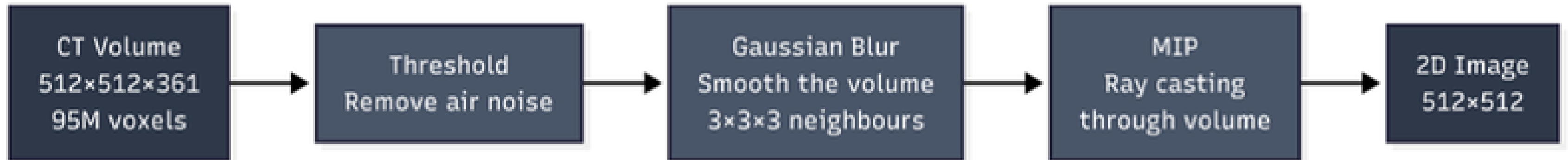


CLE - Practical Assignment 3

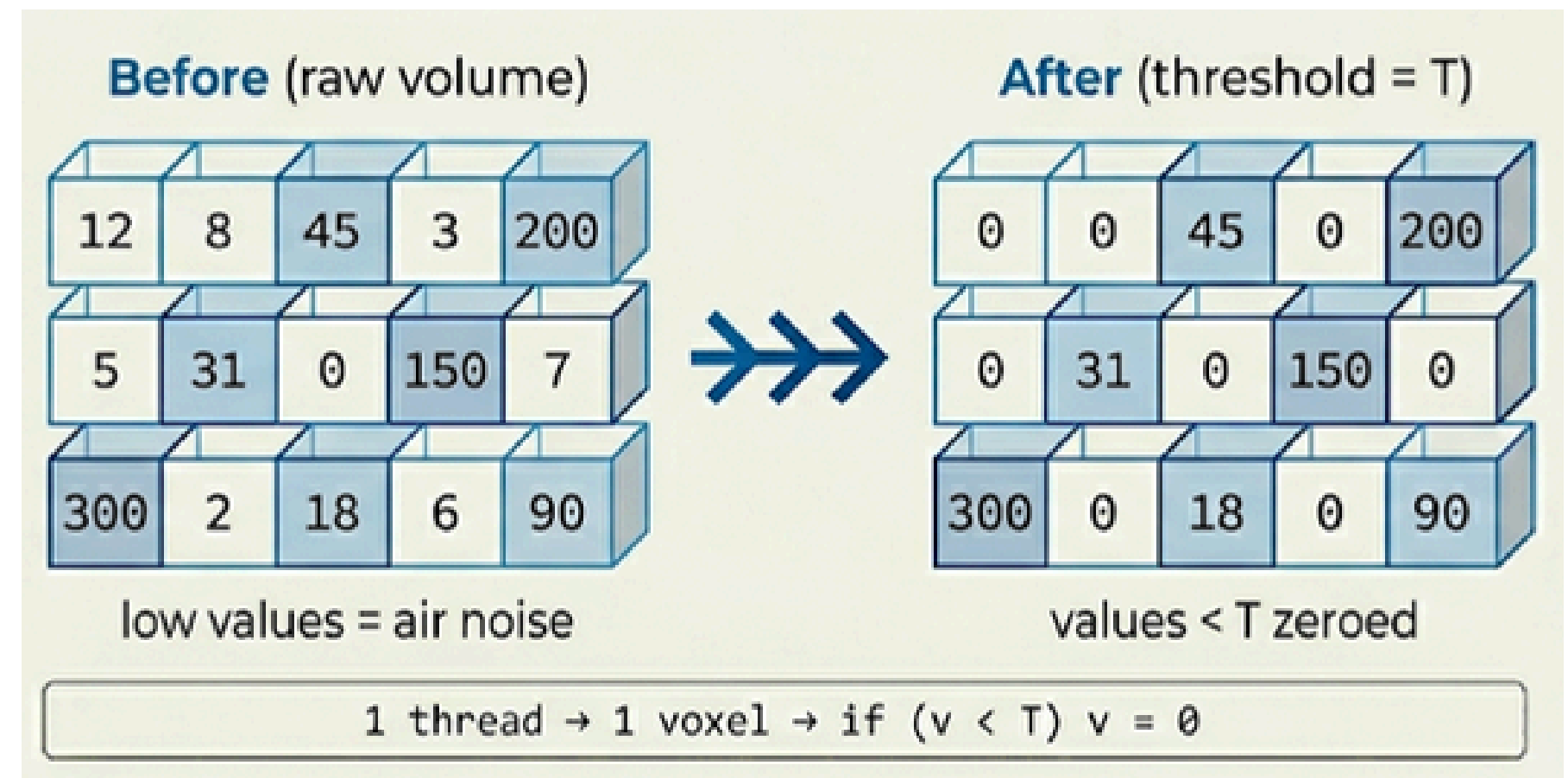
Presented by
David Pelicano (113391)
Guilherme Siva (131143)

PIPELINE



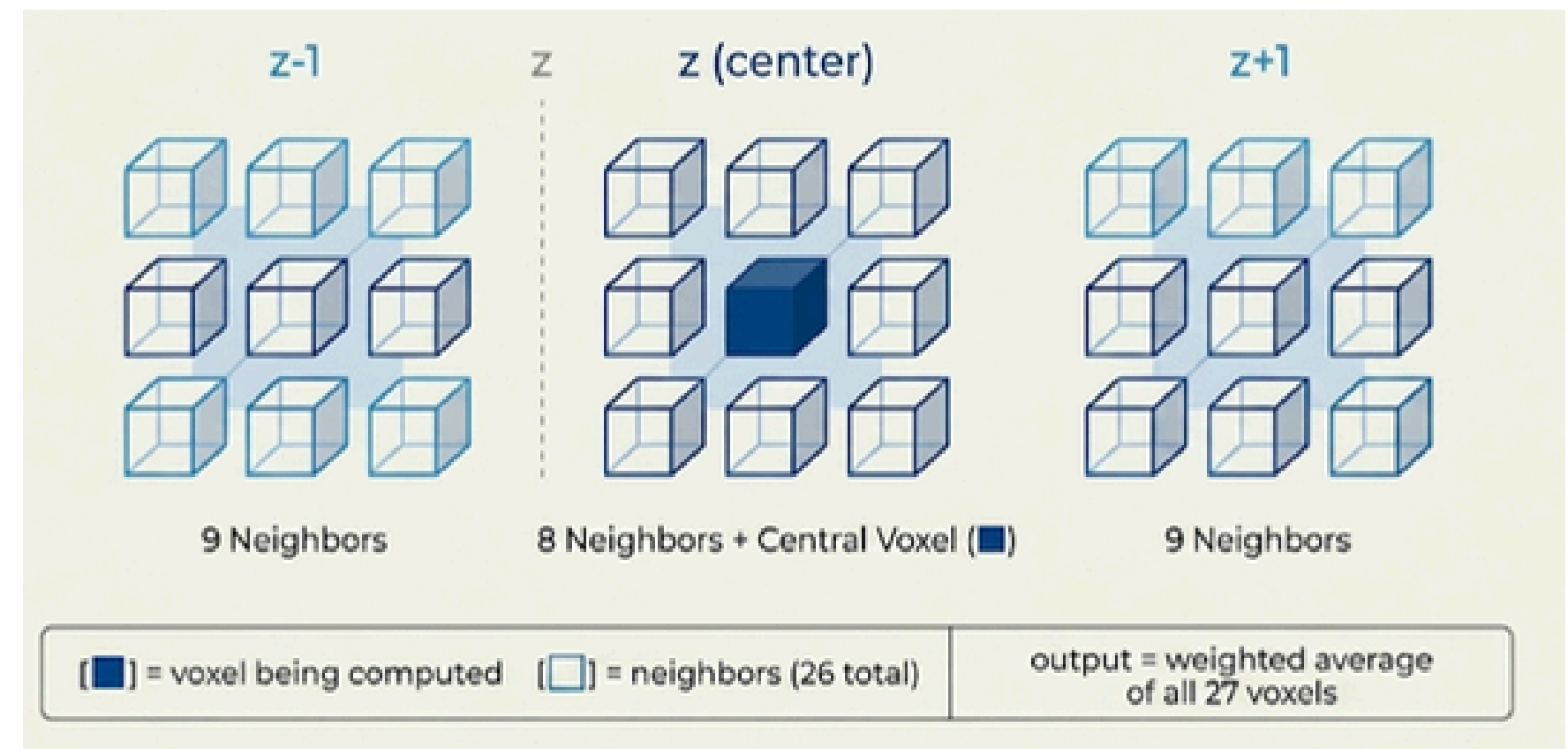
KERNEL 1: THRESHOLDING

- Volume has ~94.5 million voxels
- Low-intensity values = air noise surrounding the bunny
- **Solution:** 1 thread per voxel $\rightarrow$ zero if below threshold
- Grid: 1D, blocks of 256 threads $\rightarrow$ 369,141 blocks



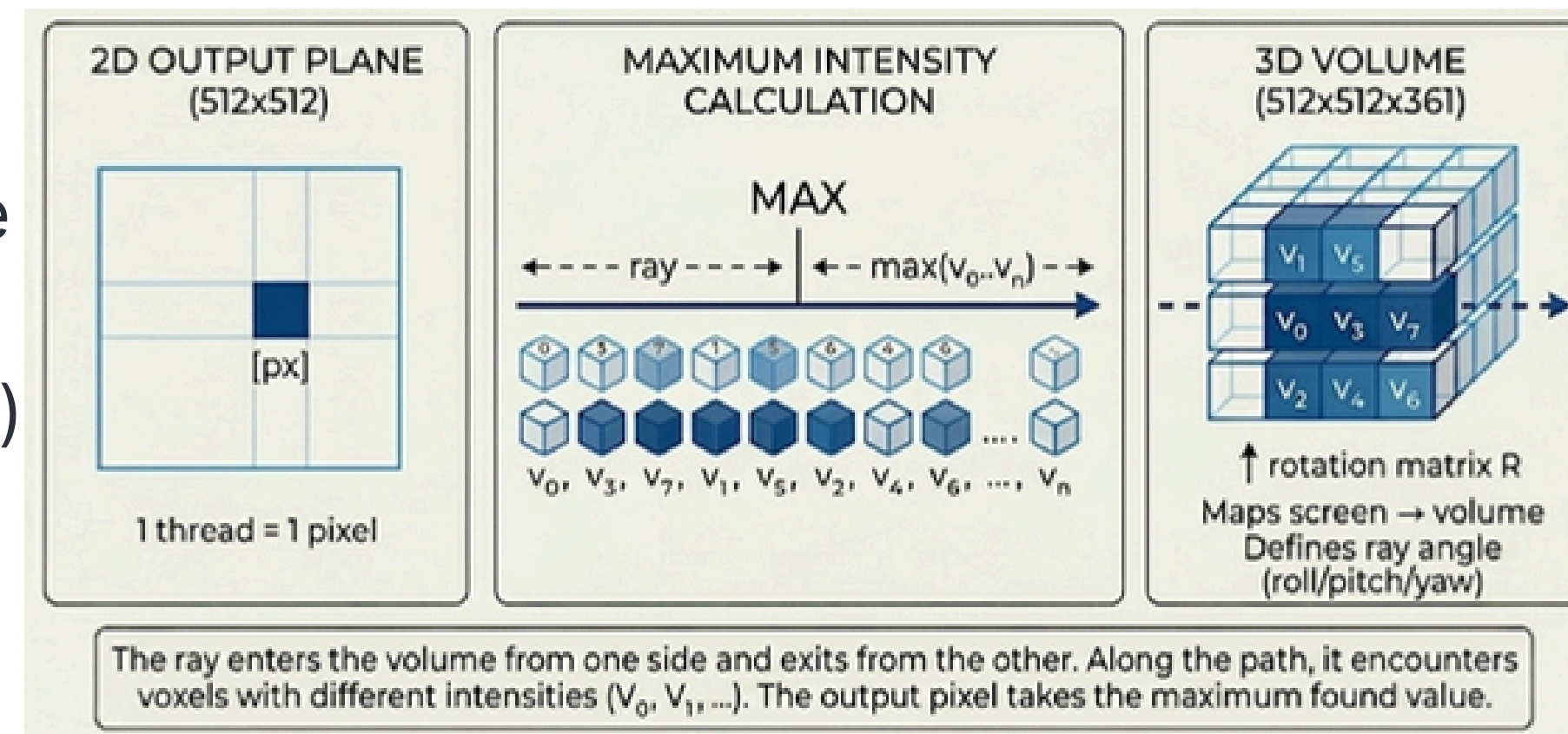
KERNEL 2: GAUSSIAN BLUR

- Prevents single high-intensity noise pixels from dominating the MIP
- Each voxel recalculated from its $3 \times 3 \times 3$ neighborhood (27 values)
- Boundary voxels: clamp to nearest edge (replicate padding)
- Grid: 3D blocks of $8 \times 8 \times 8$ threads
- Gaussian weights computed on host from σ , normalized, sent to GPU once



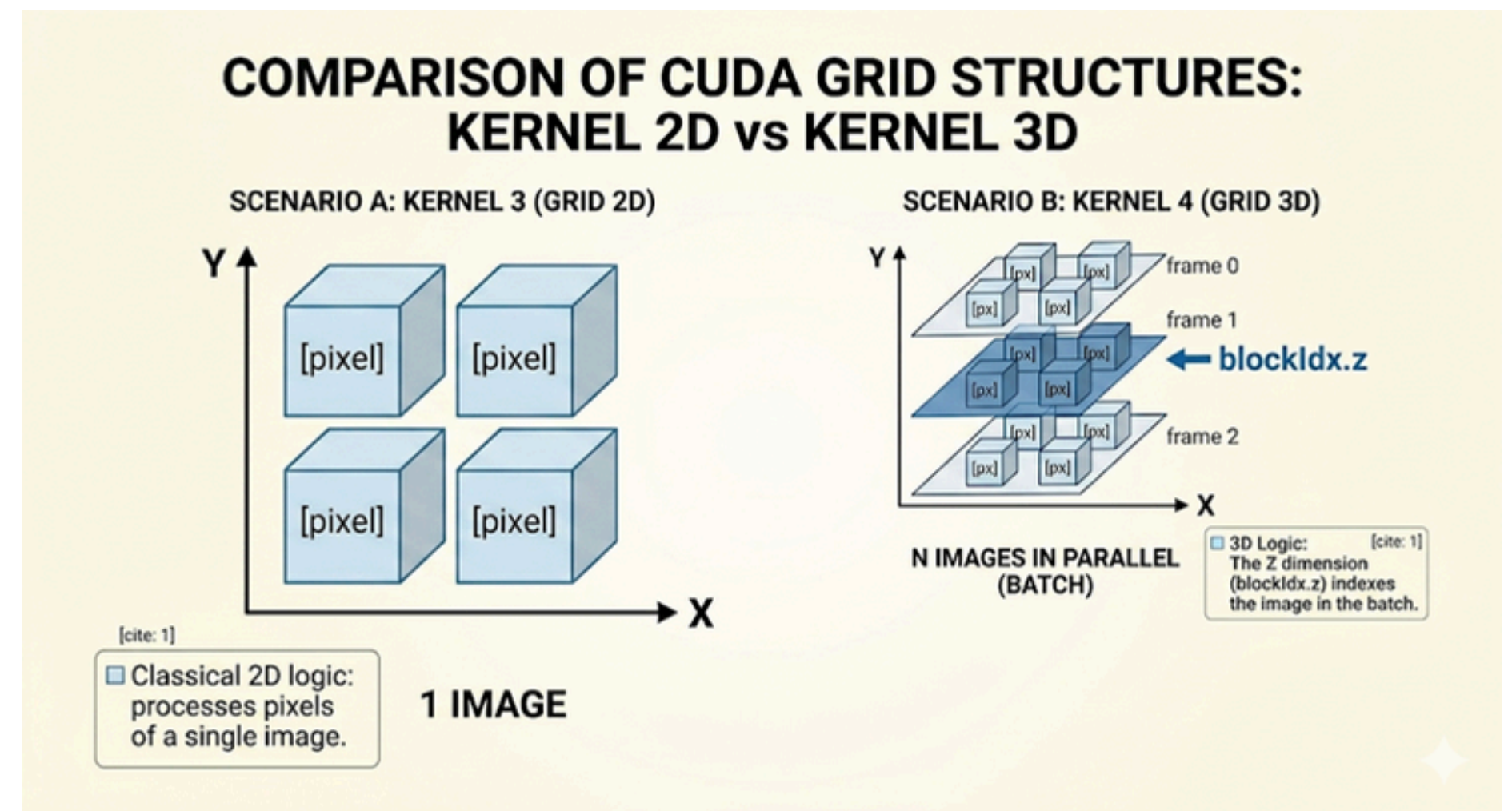
KERNEL 3: MIP + ROTATION

- 1 thread = 1 output pixel ($512 \times 512 = 262,144$ threads)
- Each thread fires a ray through the volume and keeps the maximum intensity found
- Rotation matrix R (built from roll/pitch/yaw) maps screen coordinates $\rightarrow$ volume space
- Ray length = volume diagonal $\rightarrow$ nothing gets clipped at any rotation angle
- Grid: 2D, blocks of 16×16 threads



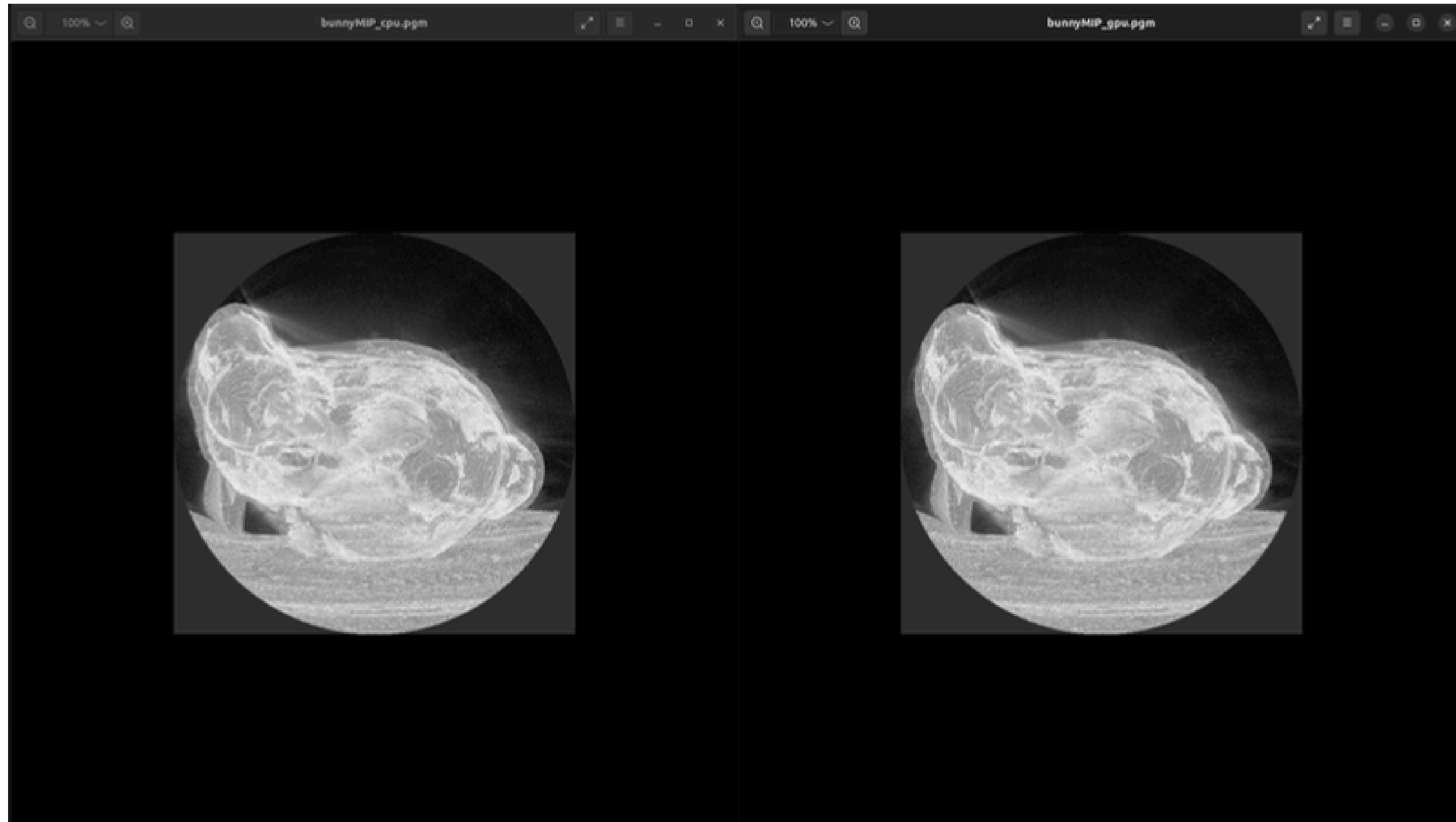
KERNEL 4: MULTI-FRAME MIP (INNOVATION)

- Extends Kernel 3 to generate N frames in a single kernel call
- Grid adds a 3rd dimension: blockIdx.z = frame index
- Each frame uses its own rotation matrix from R_all array
- All N frames computed fully in parallel - no loop on the host side
- Output: contiguous $N \times 512 \times 512$ buffer
- Used by --frames N to produce a 360° rotating video with ffmpeg



SPEEDUP ANALYSIS

Stage	CPU	GPU (kernel)	Speedup
Threshold	66 ms	2.40 ms	27x
Gaussian Blur	4200 ms	24.05 ms	175x
MIP Projection	2430 ms	1.80 ms	1350x
End-to-end	6702 ms	45 ms	148x



CLE - Practical Assignment 3

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